Sammy Suliman

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EDUCATION

New York University - Tandon School of Engineering — ***Expected Graduation: May 2027***
Electrical Engineering MS *(Specialization in Embedded Systems & Robotics)* | 3.945 GPA

University of California Santa Barbara — ***Graduated: June 2024***
Applied Mathematics / Data Science BS | 3.96 GPA

EXPERIENCE

Robotics Software Engineer — ***Aug 2024 - Dec 2025***
SINGULARITY SOLUTIONS, LLC — *Santa Barbara, CA*
- Designed a 3D model of an excavator's bucket in OnShape CAD for use in 3D simulation.
- Designed 2D simulation of excavator environment using PyGame to find the optimal path to goal using a genetic algorithm.
- Developed algorithms to automatically transform incoming LiDAR point cloud data to match predefined maps based on local curvature information with complete accuracy on training examples.
- Developed a SLAM (Simultaneous Localization & Mapping) system to graph the pose & orientation of an excavator in real time,.integrating sensor data from Velodyne LiDAR unit and IMU.

Machine Learning Intern — ***Apr 2025 - Oct 2025***
MICRON TECHNOLOGY — ***May 2026 - Aug 2026*** — *Boise, ID*
- Proposed and validated an approach for identifying faults in wafer manufacturing sensor data, using a clustering algorithm that located several anomalies missed by the existing method.
- Discovered and fixed bugs in Microsoft Azure's autoencoder time series anomaly detector (here) and pushed to the company fork for additional use in locating defective wafers.
- Created a Jupyter workflow that automatically extracted data and applied anomaly detector on wafers.

Contract Systems Software Engineer — ***Aug 2024 - Apr 2025***
VERMEER CORPORATION — *Remote*
- Developed a MQTT-based control and telemetry system with real-time UI integration and robust pause/reset handling.
- Designed a front-end interface using Kivy to enable real-time user interaction with sensor visualizations using blitting.

Machine Learning Intern — ***May 2024 - Aug 2024***
NEVADA NATIONAL SECURITY SITE — *Las Vegas, NV*
- Wrote statistical algorithms for improving radiological anomaly detection and developed ensemble combining known algorithms for improved precision, resulting in 50% reduced false positive rate.
- Developed synthetic data generating process for algorithm testing.
- Wrote functions to spatially localize streaming data into Pandas dataframes.

Modelling & Simulation Intern — ***June 2023 - Aug 2023***
PACIFIC NORTHWEST NATIONAL LABORATORIES — *Richland, WA*
- Wrote simulations for traffic flow across land border ports, designed logic to regulate lanes according to traffic.
- Created Python metrics with OOP to validate the performance of the simulation.
- Improved performance of the simulation by 10% using parallelization.

Deep Learning Intern — ***Jan 2023 - May 2023***
AMGEN, INC. — *Thousand Oaks, CA*
- Fine-tuned BERT model through TensorFlow on HPC system to identify most relevant named entities in paper abstracts from PubMed. Successfully matched 73% of genetic condition and 66% of disease named entities to correct category (Overall precision/recall 0.62/0.76 on training data).

EXTRACURRICULARS

NYU RoboMaster UltraViolet
Navigation Subteam — ***Jan 2026 - May 2026***
Computer Vision Team Lead — ***Aug 2026 - present***
- Set up ROS2 robotics workflows on Jetson/AGI platforms, including TurtleBot4 simulation, RViz visualization, ROS bag playback, RealSense camera debugging, and point-cloud display for LiDAR testing.
- Developed and tested 3D SLAM workflows with Livox LiDAR and Gazebo simulation, progressing from dependency setup and network debugging to successful 3D mapping and simulation integration.

PROJECTS

AgileNavigator | *(~100+ hours)* - GitHub Link — ***Jan 2026 - May 2026***
- Developed a remote control 2-wheeled robot that can navigate inaccessible locations and stream video of its surroundings.
- Wired battery switch to UART pins on STM32, soldered wires to PCBs to bring motors online, wired ESP32 camera to STM32.
- Wrote and fine-tuned the PID control to enable the robot to balance independently upon 2 wheels.
- Developed WiFi-based joystick control for AgileNavigator, integrated streaming video from ESP32 into the background.

zkPHIRE | *(~10+ hours)* - GitHub Link — ***Jan 2026 - May 2026***
- Built zkPHIRE, an FPGA/HLS accelerator for zero-knowledge proof SumCheck operations over the BN254 field.
- Verified design correctness with Python golden models, C++ testbenches, and synthesis on PYNQ-Z2.
- Optimized hardware using AXI, BRAM/m_axi, loop unrolling, array partitioning, and multi-PE parallelism.

BeamHear | *(~50+ hours)* - GitHub Link — ***Jan 2026 - May 2026***
- Designed an STM32-based assistive listening device that streams audio and applies real-time DSP to suppress non-speech background noise and enhance clarity for users with auditory processing disorders.

Embedded Parkinson's Detector | *(~15 hours)* - GitHub Link — ***Nov 2025 - Dec 2025***
- Designed embedded firmware on the STM32 B-L475 for a motion-sensing system that captures 52 Hz IMU data, performs real-time FFT/FIR feature extraction, and transmits Parkinson's-related motion events via interrupt-driven BLE.

Immersion.AI | *(~40 hours)* - GitHub Link — ***Jan 2025 - May 2025***
- Developed voice-based chatbot to immerse language learners in 2nd language through code-switching.
- Fine-tuned mBART on custom code-switched datasets through AWS server, integrated with SpeechRecognition API.

Direct Shear Device | *(~15 hours)* - GitHub Link — ***Dec 2024 - Jan 2025***
- Built direct shear device using Arduino Uno, motor driver, linear actuator, and power supply to test soil quality.

The EM Algorithm in Information Geometry | Senior Thesis | *(~100 hours)* - Link — ***Jan 2024 - Mar 2024***
- Was the first to implement a novel algorithm for natural gradient descent from a theoretical paper (here).

SKILLS

Languages: ***Proficient:*** Python (5yrs) · C++ (2yrs) ***Intermediate:*** SQL (4yrs) · Matlab (2yrs) ***Beginner:*** SystemVerilog (1yr)
Software: AWS · GCP · Docker · Flask · TensorFlow · PyTorch · PySpark · Kivy · ROS2 · SUMO · Git · Linux
Hardware: Cadence Virtuoso · Microcontrollers (Arduino/STM32/ESP32) · Soldering · 3D Design (OnShape CAD) · FPGAs (Pynq-Z2)